Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The differential equation of family of curves whose tangent form an angle of $\pi/4$ with the hyperbola $xy = C^2$ is

(A) $\frac{dy}{dx} = \frac{x^2+C^2}{x^2-C^2}$

(B) $\frac{dy}{dx} = \frac{x^2-C^2}{x^2+C^2}$

(C) $\frac{dy}{dx} = -\frac{C^2}{x^2}$

(D) None of these

Q.2

What is the solution of $\frac{dy}{dx} + 2y = 1$ satisfying $y(0) = 0$?

(A) $y = \frac{1-e^{-2x}}{2}$

(B) $y = \frac{1+e^{-2x}}{2}$

(C) $y = 1 + e^x$

(D) $y = \frac{1+e^x}{2}$

Q.3

What is the solution of the differential equation
$\frac{dy}{dx} = \frac{y}{(x+2y^3)}$?

(A) $y(1 - xy) = cx$

(B) $y^3 - x = cy$

(C) $x(1 - xy) = cy$

(D) $x(1 + xy) = cy$

Q.4

A differential equation associated with the primitive $y = a + be^{5x} + ce^{-7x}$ is

(A) $y_3 + 2y_2 - y_1 = 0$

(B) $y_3 + 2y_2 - 35y_1 = 0$

(C) $4y_3 + 5y_2 - 20y_1 = 0$

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(D) None of these

Q.5

The differential equations of all conics whose axes coincide with the co-ordinate axis

(A) $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2+y\frac{dy}{dx}=0$

(B) $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2+x\frac{dy}{dx}=0$

(C) $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

(D) $xy\frac{d^2y}{dx^2}-x\left(\frac{dy}{dx}\right)^2+y\frac{dy}{dx}=0$

Q.6

if $y^2=p(x)$ is a polynomial of degree 3, then what is $2\frac{\mathrm{d}}{\mathrm{dx}}\left[\mathrm{y}^3\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dx}^2}\right]$ equal to ?

(A) $p'(x)p'''(x)$

(B) $\mathrm{p''(x)p'''(x)}$

(C) $\mathrm{p(x)p'''(x)}$

(D) A constant

Q.7

If $\mathrm{y}=\left(x+\sqrt{1+x^2}\right)^n$, then $\left(1+x^2\right)\frac{d^2y}{dx^2}+x\frac{dy}{dx}$ is

(A) $\mathrm{n^2y}$

(B) $-n^2y$

(C) -y

(D) $2x^2y$

Q.8

If $\mathrm{y}=\mathrm{e}^{4\mathrm{x}}+2\mathrm{e}^{-\mathrm{x}}$ satisfies the relation $\frac{\mathrm{d}^3\mathrm{y}}{\mathrm{dx}^3}+\mathrm{A}\frac{\mathrm{dy}}{\mathrm{dx}}+\mathrm{By}=0$, then values of A and B respectively are :

(A) $-13,14$

(B) $-13,-12$

(C) $-13,12$

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(D) $12, -13$

Q.9

Solution of the differential equation

$x = 1 + xy\frac{dy}{dx} + \frac{x^2y^2}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{x^3y^3}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots\ldots\ldots\ldots$

(A) $y = \ln(x) + c$

(B) $y = (\ln x)^2 + c$

(C) $y = \pm\ln(x) + c$

(D) $xy = x^y + c$

Q.10

$y = 2\cos x + 3\sin x$ satisfies which of the following differential equations?

1 $\frac{d^2y}{dx^2} + y = 0$

2 $\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} = 0$

Select the correct answer using the code given below.

(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2

Q.11

The equation of the curve such that the subtangent at any point of the curve is two times the abscissa of the point and curve passes through point $1, 2$ is

(A) $y^2 = x + 3$

(B) $y = x^2$

(C) $y^2 = 4x$

(D) $y = 2x^2$

Q.12

A curve is such that the portion of the x -axis cut off between the origin and the tangent at a point is twice the abscissa and which passes through the point $(1, 2)$. The equation of the curve is

(A) $xy = 1$

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(B) $xy = 2$

(C) $xy = 3$

(D) None of these

Q.13

What is the differential equation for $y^2 = 4a(x - a)$?

(A) $yy' - 2xyy' + y^2 = 0$

(B) $yy'(yy' + 2x) + y^2 = 0$

(C) $yy'(yy' - 2x) + y^2 = 0$

(D) $yy' - 2xyy' + y = 0$

Q.14

The differential equation of the curve
$\frac{x}{c-1} + \frac{y}{c+1} = 1$ is given by

(A) $\left(\frac{dy}{dx} - 1\right)\left(y + x\frac{dy}{dx}\right) = 2\frac{dy}{dx}$

(B) $\left(\frac{dy}{dx} + 1\right)\left(y - x\frac{dy}{dx}\right) = \frac{dy}{dx}$

(C) $\left(\frac{dy}{dx} + 1\right)\left(y - x\frac{dy}{dx}\right) = 2\frac{dy}{dx}$

(D) None of these

Q.15

If $y = e^{-x}(A\cos x + B\sin x)$, then y satisfies

(A) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$

(B) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$

(C) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$

(D) $\frac{d^2y}{dx^2} + 2y = 0$

Q.16

If $y = \left(x + \sqrt{1 + x^2}\right)^n$, then $\left(1 + x^2\right)\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dx}^2} + \mathrm{x}\frac{\mathrm{dy}}{\mathrm{dx}}$ is

(A) n^2y

(B) $-n^2y$

(C) $-y$

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(D) $2x^2y$

Q.17

The differential equations of all conics whose axes coincide with the co-ordinate axis

(A) $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2+y\frac{dy}{dx}=0$

(B) $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2+x\frac{dy}{dx}=0$

(C) $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

(D) $xy\frac{d^2y}{dx^2}-x\left(\frac{dy}{dx}\right)^2+y\frac{dy}{dx}=0$

Q.18

If for the differential equation $y'=\frac{y}{x}+\phi\left(\frac{x}{y}\right)$, the general solution is $y=\frac{x}{\log|Cx|}$, then $\phi(x/y)$ is given by

(A) $-x^2/y^2$

(B) y^2/x^2

(C) x^2/y^2

(D) $-y^2/x^2$

Q.19

The line normal to a given curve at each point (x,y) on the curve passes through the point $(3,0)$. If the curve contains the point $(3,4)$ then its equation is

(A) $x^2+y^2+6x-7=0$

(B) $2\left(\mathrm{x}^2+\mathrm{y}^2\right)-12\mathrm{x}-7=0$

(C) $x^2+y^2-6x-7=0$

(D) None of these

Q.20

Differential equation of all conics of the form $ax^2+by^2=1$ and b being parameters is :

(A) $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

(B) $xy\frac{d^2y}{dx^2}-x\left(\frac{dy}{dx}\right)^2+y\frac{dy}{dx}=0$

(C) $xy\frac{d^2y}{dx^2}-x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

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(D) None of these

Answers & Solutions

Q.1 Answer:

$\frac{dy}{dx} = \frac{x^2-C^2}{x^2+C^2}$

Solution:

Let the slope of tangent of required family be $\frac{dy}{dx} = \mathrm{m}_1$

Also $y = \frac{C^2}{x}$**; therefore,** $\frac{dy}{dx} = -\frac{C^2}{x^2} = \mathrm{m}^2$ **(say).**

By the given condition, we have $\tan\frac{\pi}{4} = \frac{m_1-m_2}{1+m_1m_2} \Rightarrow 1 + \mathrm{m}_1\,\mathrm{m}_2 = \mathrm{m}_1 - \mathrm{m}_2$

$\Rightarrow \frac{dy}{dx} + \frac{C^2}{x^2} = 1 - \frac{C^2}{x^2}\frac{dy}{dx} \Rightarrow \frac{dy}{dx}\left(1 + \frac{C^2}{x^2}\right)$

$= 1 - \frac{C^2}{x^2} \Rightarrow \frac{dy}{dx} = \frac{x^2-C^2}{x^2+C^2}$

Q.2 Answer:

$y = \frac{1-e^{-2x}}{2}$

Solution:

$\frac{dy}{dx} + 2y = 1; \frac{dy}{dx} = 1 - 2y$

$\int \frac{\mathrm{dy}}{1-2\mathrm{y}} = \int \mathrm{dx}$

$-\frac{1}{2}\log|1-2y| = x + C$

at $x = 0, y = 0$

$-\frac{1}{2}\log 1 = 0 + \mathrm{C} \Rightarrow \mathrm{C} = 0$

$1 - 2y = e^{-2x}; y = \frac{1-e^{-2x}}{2}$

Q.3 Answer:

$y^3 - x = cy$

Solution:

$\mathrm{y}^3 - \mathrm{x} = \mathrm{cy}$

$\Rightarrow \quad 3y^2\frac{dy}{dx} - 1 = c\frac{dy}{dx} \Rightarrow \frac{dy}{dx}\left(3y^2 - c\right) = 1$

$\Rightarrow \quad \frac{dy}{dx}\left(3y^2 - \frac{y^3-x}{y}\right) = 1$

$\Rightarrow \frac{dy}{dx}\left(\frac{3y^3-y^3+x}{y}\right) = 1 \Rightarrow \frac{dy}{dx} = \frac{y}{x+2y^3}$

Q.4 Answer:

$y_3 + 2y_2 - 35y_1 = 0$

Solution:

$y = a + be^{5x} + ce^{-7x}$

$\therefore \quad y_1 = 0 + 5be^{5x} - 7ce^{-7x}$

Dividing by ye^{5x}**, we get :** $e^{-5x}y_1 = 5b - 7ce^{-12x}$ **Again differentiating both sides w.r.t.** x**, we get**

$e^{-5x}\cdot y_2 + y_1(-5)e^{-5x} = 0 + 84ce^{-12x}$

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Dividing by e^{-12x}, we get : $e^{7x}\left(y_2-5y_1\right)=84c$ Differentiating both sides w.r.t. x, we get

$e^{7x}\left(y_3-5y_2\right)+\left(y_2-5y_1\right).7e^{7x}=0$

$\Rightarrow y_3+2y_2-35y_1=0$

Q.5 Answer:

$xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

Solution:

Any conic whose axes coincide with coordinate axis is $ax^2+by^2=1$

Diff. both sides w.r.t. ' x ', we get $2ax+2by\frac{dy}{dx}=0$ i.e. $ax+by\frac{dy}{dx}=0$

Diff. again, $a+b\left(y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right)=0$

From

(ii), $\frac{a}{b}=-\frac{ydy/dx}{x}$

From (iii), $\frac{a}{b}=-\left(y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right)$

$\therefore \quad \frac{y\frac{dy}{dx}}{x}=y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2$

$\Rightarrow \quad xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

Q.6 Answer:

$\mathrm{p(x)p'''(x)}$

Solution:

Given that $y^2=p(x)$ Differentiating $\Rightarrow$ $\quad 2\mathrm{yy_1}=\mathrm{p'(x)}\quad$ **[here** $\mathrm{y_1}=\frac{\mathrm{dy}}{\mathrm{dx}}$

$\Rightarrow \quad 2\mathrm{y_1}=\frac{\mathrm{p'(x)}}{\mathrm{y}}$

Differentiating again, $\Rightarrow \quad 2\mathrm{y_2}=\frac{\mathrm{yp''(x)-p'(x)y_1}}{\mathrm{y^2}},\left[\mathrm{y_2}=\frac{\mathrm{d^2y}}{\mathrm{dx^2}}\right]$

$\Rightarrow 2\mathrm{y_2}=\frac{\mathrm{yp''(x)}-\frac{\mathrm{p'(x)\cdot p'(x)}}{2\mathrm{y}}}{\mathrm{y^2}}$

$=\frac{2y^2p''(x)-p'(x))^2}{2y^3}$

$\Rightarrow \quad 2\mathrm{y^3y_2}=\frac{1}{2}\left[2\mathrm{y^2p''(x)}-\mathrm{(p'(x))^2}\right]$

$\Rightarrow \quad 2y^3y_2=\frac{1}{2}\left[2p(x)p''(x)-(p'(x))^2\right]$

$\Rightarrow \quad 2\frac{\mathrm{d}}{\mathrm{dx}}\left(\mathrm{y^3y_2}\right)$

$=\frac{1}{2}\left[2p'(x)p''(x)+2p(x)p'''(x)-2p'(x)p''(x)\right]$

$=\mathrm{p'(x)p'''(x)}$

Q.7 Answer:

$\mathrm{n^2y}$

Solution:

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$y = \left(x + \sqrt{1+x^2}\right)^n$

$\frac{dy}{dx} = n\left(x + \sqrt{1+x^2}\right)^{n-1}\left(1 + \frac{1}{2}\left(1+x^2\right)^{-1/2} \cdot 2x\right)$

$\frac{dy}{dx} = n\left(x + \sqrt{1+x^2}\right)^{n-1}\frac{\left(\sqrt{1+x^2}+x\right)}{\sqrt{1+x^2}} = \frac{n\left(\sqrt{1+x^2}+x\right)^n}{\sqrt{1+x^2}}$

or $\sqrt{1+x^2}\frac{dy}{dx} = ny$ **or** $\sqrt{1+x^2}y_1 = ny \quad \left(y_1 = \frac{dy}{dx}\right)$

Squaring, $\left(1+x^2\right)y_1^2 = n^2y^2$

Differentiating, $\left(1+x^2\right)2y_1y_2 + y_1^2 \cdot 2x = n^2 \cdot 2yy_1$

$\left(\right.$ **Here,** $y_2 = \frac{d^2y}{dx^2}\left.\right)$ **or** $\left(1+x^2\right)y_2 + xy_1 = n^2y$

Q.8 Answer:

$-13, -12$

Solution:

Given $y = e^{4x} + 2e^{-x}$ **Differentiating we get** $\frac{dy}{dx} = 4e^{4x} - 2e^{-x} \Rightarrow \frac{d^2y}{dx^2} = 16e^{4x} + 2e^{-x}$

$\Rightarrow \frac{\mathrm{d}^3\mathrm{y}}{\mathrm{dx}^3} = 64\mathrm{e}^{4\mathrm{x}} - 2\mathrm{e}^{-\mathrm{x}}$

Putting these values in $\frac{\mathrm{d}^3\mathrm{y}}{\mathrm{dx}^3} + \mathrm{A}\frac{\mathrm{dy}}{\mathrm{dx}} + \mathrm{By} = 0$

We have,

$(64 + 4\,\mathrm{A} + \mathrm{B})\mathrm{e}^{4\mathrm{x}} + (-2 - 2\,\mathrm{A} + 2\,\mathrm{B})\mathrm{e}^{-\mathrm{x}} = 0$

$\Rightarrow 64 + 4\,\mathrm{A} + \mathrm{B} = 0, -2 - 2\,\mathrm{A} + 2\,\mathrm{B} = 0$

Solving these eqs., we get $\mathrm{A} = -13,\ \mathrm{B} = -12$

Q.9 Answer:

$y = \pm\ln(x) + c$

Solution:

The given equation is reduced to $x = e^{xy(dy/dx)}$

$\Rightarrow \quad \ln x = xy\frac{dy}{dx} \Rightarrow \int ydy = \int \frac{1}{x}\ln x dx$

$\Rightarrow \quad \frac{y^2}{2} = \frac{(\ell nx)^2}{2} + c$

$\Rightarrow \quad y = \pm\sqrt{(\ln x)^2} + c = \pm\ln x + c$

Q.10 Answer:

1 only

Solution:

$y = 2\cos x + 3\sin x$

$\Longrightarrow \frac{dy}{dx} = -2\sin x + 3\cos x$

$\Longrightarrow \frac{d^2y}{dx^2} = -2\cos x - 3\sin x$

We can see that,

$\mathrm{y} + \frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dx}^2} = 0$

Thus, statement 1 is true.

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Also, $\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} = 0$

Thus, statement 2 is not true.

Hence, option A is correct.

Q.11 Answer:

$y^2 = 4x$

Solution:

Subtangent $\frac{y}{m} = 2x$

$\Rightarrow 2xm = y \Rightarrow 2x\frac{dy}{dx} = y \Rightarrow 2\frac{dy}{y} = \frac{dx}{x}$

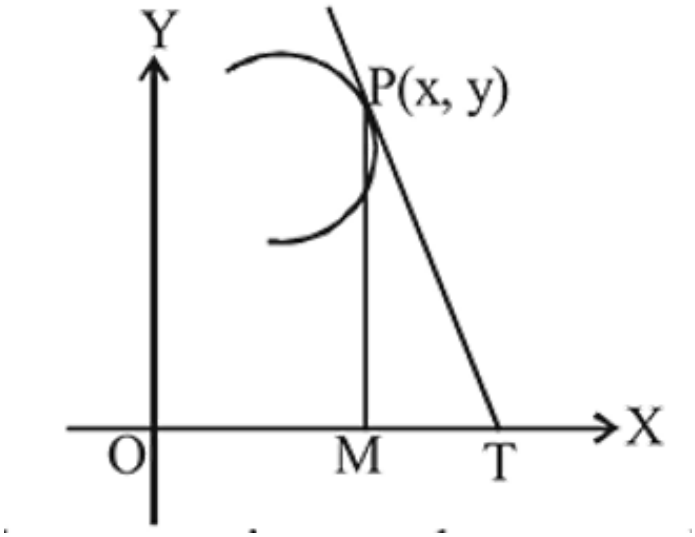
on integrating, $2lny = lnx + lnc \Rightarrow y^2 = cx$ **passes through** $1, 2$**, so**

$\therefore \quad y^2 = 4x$

Q.12 Answer:

$xy = 2$

Solution:



Let $\mathrm{P(x, y)}$ **be any point on the curve,** PM **the perpendicular to x-axis PT the tangent at** P **meetingthe axis of** x **at** T**. As given** $OT = 2$ **OM** $= 2x$**. Equation of the tangent at** $P(x, y)$ **is**

$Y - y = \frac{dy}{dx}(X - x)$

It intersects the axis of x **where** $Y = 0$

i.e. $-\mathrm{y} = \frac{\mathrm{dy}}{\mathrm{dx}}(\mathrm{X} - \mathrm{x})$ **or** $\mathrm{X} = \mathrm{x} - \mathrm{y}\frac{\mathrm{dy}}{\mathrm{dx}} = \mathrm{OT}$

Hence $x - y\frac{dy}{dx} = 2x$ **or** $\frac{dx}{x} + \frac{dy}{y} = 0$

Integrating, $\log x + \log y = \log C$ **i.e.** $xy = C$

This passes through $(1, 2), \quad \therefore \mathrm{C} = 2$**. Hence the required curve is** $\mathrm{xy} = 2$**.**

Q.13 Answer:

$yy'\,(yy' - 2x) + y^2 = 0$

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Solution:

Given curve is $y^2 = 4a(x-a)$ On differentiating w.r.t. x, we get $2yy' = 4a$

$\Rightarrow \quad a = \frac{yy'}{2}$

On putting the value of a in Eq. (i), we get $y^2 = 4\left(\frac{yy'}{2}\right)\left(x - \frac{yy'}{2}\right) = yy'(2x - yy')$

$\Rightarrow yy'(yy' - 2x) + y^2 = 0$

Q.14 Answer:

$\left(\frac{dy}{dx} + 1\right)\left(y - x\frac{dy}{dx}\right) = 2\frac{dy}{dx}$

Solution:

$\frac{x}{c-1} + \frac{y}{c+1} = 1$

$\Rightarrow \quad \frac{1}{c-1} + \frac{y'}{c+1} = 0$

$\Rightarrow \quad \frac{y'}{1} = \frac{c+1}{1-c} \Rightarrow \frac{y'-1}{y'+1} = c$

Put value of c in (1)

$\Rightarrow \frac{x}{\frac{y'-1}{y'+1}-1} + \frac{y}{\frac{y'-1}{y'+1}+1} = 1$

$\Rightarrow \quad \frac{x\left(y'+1\right)}{-2} + \frac{y\left(y'+1\right)}{2y'} = 1$

$\Rightarrow \quad \frac{\left(y'+1\right)}{2}\left(\frac{y}{y'} - x\right) = 1$

$\Rightarrow \left(1 + \frac{dy}{dx}\right)\left(y - x\frac{dy}{dx}\right) = 2\frac{dy}{dx}$

Q.15 Answer:

$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$

Solution:

$y = e^{-x}(A\cos x + B\sin x)$

$\Rightarrow \frac{dy}{dx} = e^{-x}(-A\sin x + B\cos x) - e^{-x}(A\cos x + B\sin x)$

$\Rightarrow \frac{dy}{dx} = e^{-x}(-A\sin x + B\cos x) - y$

$\ldots\ldots\ldots(2)$

$\Rightarrow \frac{d^2y}{dx^2} = e^{-x}(-A\cos x - B\sin x) - e^{-x}(-A\sin x + B\cos x)\left(-\frac{dy}{dx}\right)$

Using (1) and (2), we get

$\frac{d^2y}{dx^2} + \frac{dy}{dx} = -y - y - \frac{dy}{dx} \Rightarrow (c)$

Q.16 Answer:

n^2y

Solution:

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$$y = \left(x + \sqrt{1+x^2}\right)^n$$
$$\frac{dy}{dx} = n\left(x + \sqrt{1+x^2}\right)^{n-1}\left(1 + \frac{1}{2}\left(1+x^2\right)^{-1/2} \cdot 2x\right)$$
$$\frac{dy}{dx} = n\left(x + \sqrt{1+x^2}\right)^{n-1} \frac{\left(\sqrt{1+x^2}+x\right)}{\sqrt{1+x^2}} = \frac{n\left(\sqrt{1+x^2}+x\right)^n}{\sqrt{1+x^2}}$$
$$\text{or } \sqrt{1+x^2}\frac{dy}{dx} = ny \text{ or } \sqrt{1+x^2}y_1 = ny \quad \left(\because y_1 = \frac{dy}{dx}\right)$$
$$\text{Squaring, } \left(1+x^2\right)y_1^2 = n^2y^2$$
$$\text{Differentiating, } \left(1+x^2\right)2y_1y_2 + y_1^2 \cdot 2x = n^2 \cdot 2yy_1$$
$$\left(\text{Here, } y_2 = \frac{d^2y}{dx^2}\right) \text{ or } \left(1+x^2\right)y_2 + xy_1 = n^2y$$

Q.17 Answer:

$$xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$$

Solution:

Any conic whose axes coincide with co-ordinate axis is $ax^2 + by^2 = 1$

Diff. both sides w.r.t. $'x'$**, we get**

$2ax + 2by\frac{dy}{dx} = 0$ **i.e.** $ax + by\frac{dy}{dx} = 0$

Diff. again, $a + b\left(y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right) = 0$

(ii), $\frac{a}{b} = -\frac{ydy/dx}{x}$

From

From (iii), $\frac{a}{b} = -\left(y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right)$

$$\therefore \quad \frac{y\frac{dy}{dx}}{x} = y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2$$
$$\Rightarrow \quad xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$$

Q.18 Answer:

$-y^2/x^2$

Solution:

Putting $v = y/x$ **so that** $x\frac{dv}{dx} + v = \frac{dy}{dx}$

We have $x\frac{dv}{dx} + v = v + \phi(1/v)$

$$\Rightarrow \quad \frac{dv}{\phi(1/v)} = \frac{dx}{x}$$
$$\Rightarrow \quad \log|Cx| = \int \frac{dv}{\phi(1/v)}$$

(C being constant of integration)

But $y = \frac{x}{\log|Cx|}$ **is the general solution,**

So $\frac{x}{y} = \frac{1}{v} = \log|Cx| = \int \frac{dv}{\phi(1/v)}$

$$\Rightarrow \quad \phi(1/v) = -1/v^2$$

(differentiating w.r.t. v **both sides)** $\Rightarrow \phi(x/y) = -y^2/x^2$

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Q.19 Answer:

$x^2 + y^2 - 6x - 7 = 0$

Solution:

The equation of normal to a curve at a point (x, y) is

$(Y - y)\frac{dy}{dx} + (X - x) = 0$

Since it passes throgh the point $(3, 0)$, we have

$(0 - y)\frac{dy}{dx} + (3 - x) = 0$

$\Rightarrow y\frac{dy}{dx} = (3 - x) \Rightarrow ydy = (3 - x)dx$

Integrating, we get $\frac{y^2}{2} = 3x - \frac{x^2}{2} + C \Rightarrow x^2 + y^2 - 6x - 2c = 0$

Since the curve passes through $(3, 4)$, we have $9 + 16 - 18 - 2c = 0 \Rightarrow c = \frac{7}{2}$

$\therefore x^2 + y^2 - 6x - 7 = 0$ **is the required equation of the curve.**

Q.20 Answer:

$xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$

Solution:

The given equation is $ax^2 + by^2 = 1$.

Differentiating we get, $2ax + 2$ **by** $\frac{dy}{dx} = 0$

$\Rightarrow ax +$ **by** $\frac{dy}{dx} = 0$

Differentiating again, $a + b\left(y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right) = 0, \ldots$ **(2)**

From eqs. (1) and (2), we get $a = -\frac{by}{x}\frac{dy}{dx} = -b\left(y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right)$

$\Rightarrow xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$

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